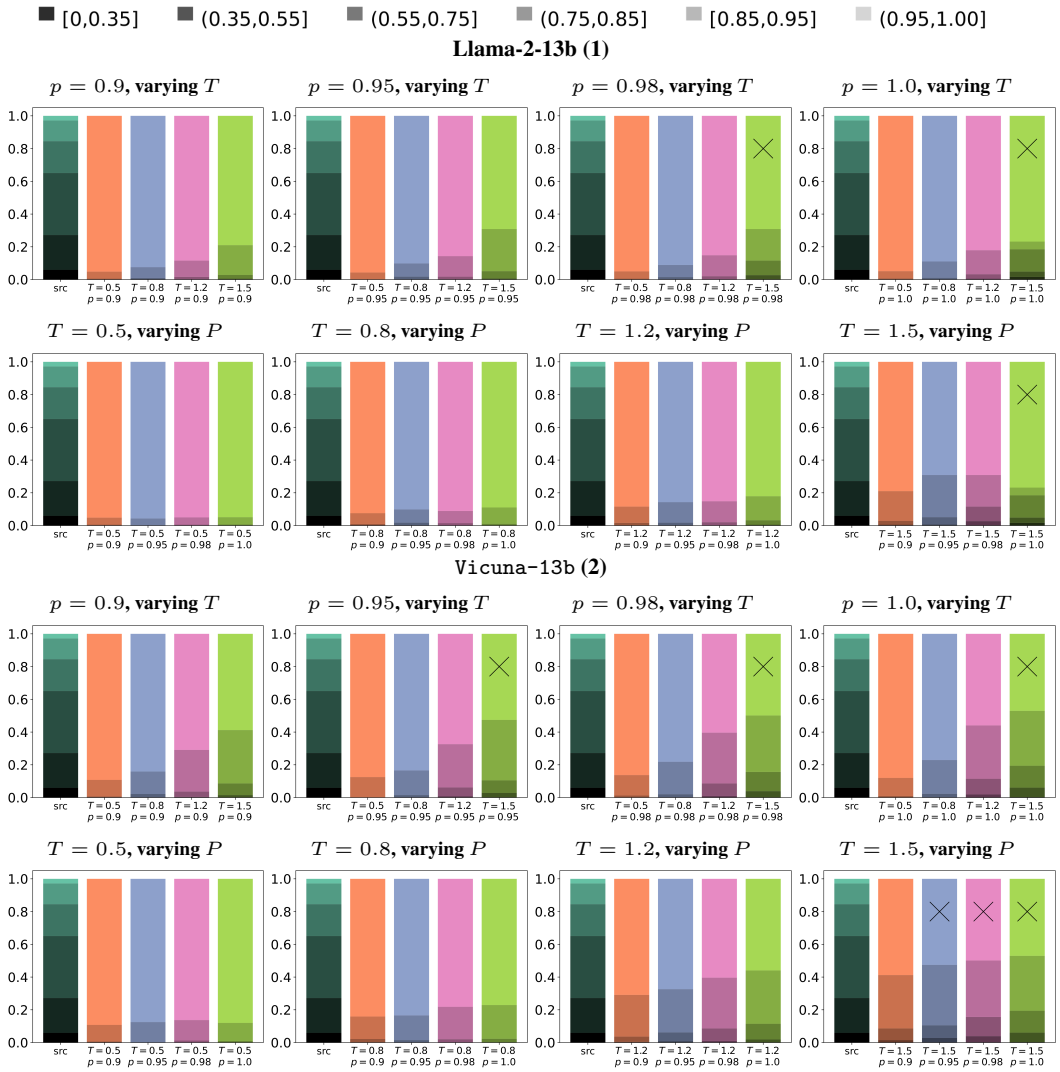

Figure 6: **Mean sentiment scores** across different models (`Llama-2-chat`, `Vicuna-13b`) and prompts ((1) and (2)), and varying sampling parameters (temperature T and top-p). We do observe an increase of diversity along the increase of generation randomness (i.e., increasing T and p), but the gap remains high between the source and even the highest generation randomness setting. Moreover, we put a black cross $\times$ on the top of the setting with a concerning low number of valid number of generations (see Table 1 for a concrete explanation); they mostly only occur for the few highest randomness settings.

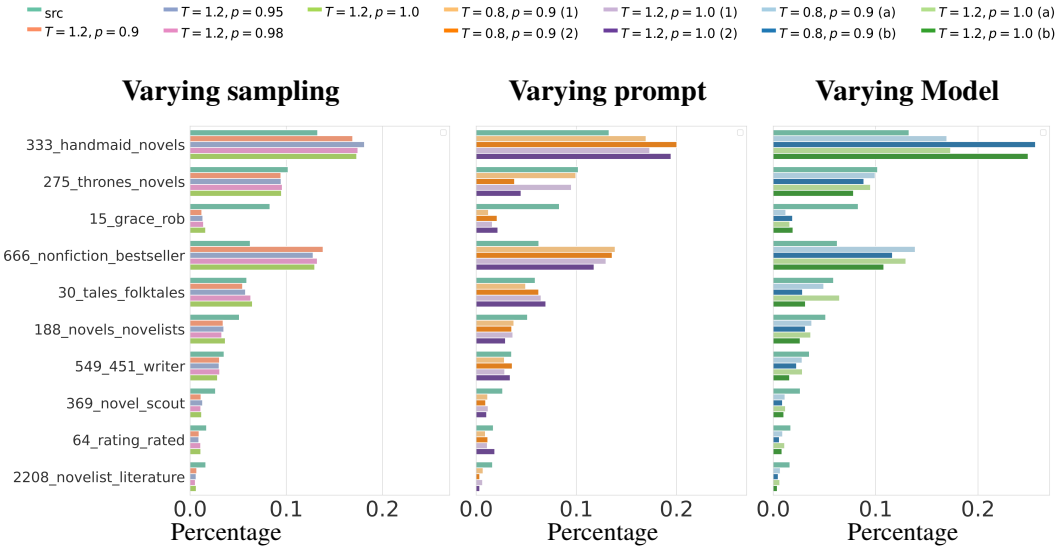

Figure 7: **Grouped bar charts of the unconditional distribution of the topic** (i.e., the distribution of the topic over all the samples). **(Left)**: generated by `Llama-2-chat` at different generation kwargs, where we fixed $T = 1.2$ and varied $p \in \{0.90, 0.95, 0.98, 1.00\}$. The prompt used is (1) as introduced in Section 6.1. **(Middle)**: generated by `Llama-2-chat` under different prompts (1) and (2). **(Right)**: generated by two models (a) `Llama-2-chat` and (b) `Vicuna-13b`, using the prompt (1). Across all three subfigures, we observe highly similar distribution shift signified by the over- and under-representation of certain topics.

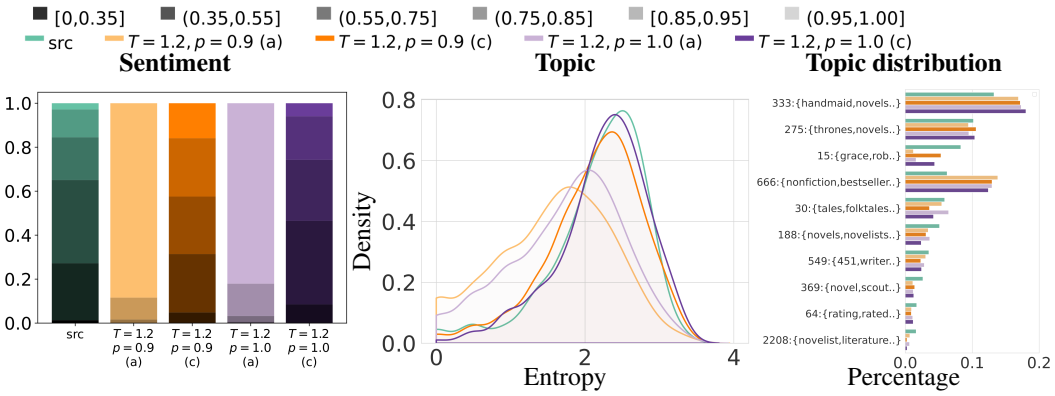

Figure 8: **Comparison between (a) the *aligned* model `Llama-2-chat` and (c) the *pre-trained* model `Llama-2`**, under prompt (1), temperature $T = 1.2$, and varying top-p parameters. We present a subset of the results (sentiment, topic, and topic distribution), corresponding to Figure 7. The figures show that the pre-trained model enjoys much better diversity than the aligned model, and is much closer to the source.

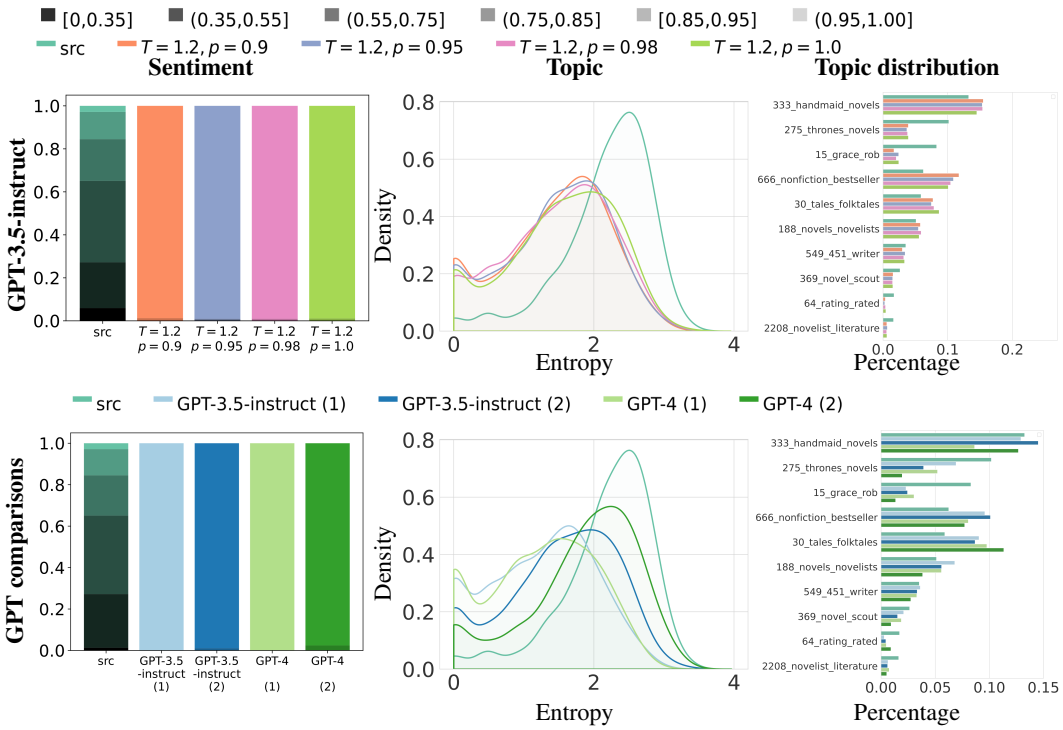

Figure 9: **Plots for OpenAI models**. **(Upper row)**: `GPT-3.5` under prompt (1), $T = 1.2$, and varing p. **(Lower row)**: comparing `GPT-3.5` and `GPT-4` at $T = 1.2, p = 1.0$ and the two propmts (1) and (2). We present a subset of the results (sentiment, topic, and topic distribution), corresponding to Figure 7. The figures show that both `GPT-3.5` and `GPT-4` display severe GeMo, which cannot be effectively mitigated via varying the sampling parameters or using a diversity-inducing prompt.

The sentiment results (in column 1) suggest that both models are overwhelmingly and invariably positive in their reviews, across various sampling parameters and prompts.

The results of the conditional topic distribution (in column 2) reveal similar conclusions—there is a significant deviation between the generated reviews and the source reviews across varying sampling parameters (see upper low) and for both GPT-3.5-instruct and GPT-4 (see lower row). Nevertheless, using a diversity-inducing prompt, i.e., prompt (2) does lead to increased diversity (see lower row).

The results of the unconditional topic distribution (in column 3) also demonstrate the distribution shift, though in a slightly different way than observed for llama family models (see presented in Appendix C.3). Concretely, the topic group 333 does not experience a significantly over-representation, while the topic group 30 does. As discussed previously, we attribute the difference in the nuanced topic distribution mainly to the difference in the training data.

C.6 Mitigation via temperature decay

We describe the decaying temperature scheme. Concretely, we choose the starting temperature $T = 10.0$ and follow a linear schedule for temperature decay, over the course of 50 timesteps (i.e., from the 1-st output token to the 50-th output token), with an ending temperature $T = 1.2$. The method is inspired by Carlini et al. [28] which reported this scheme as one sampling method that induce diversity and high quality output[1].

We present the results in Fig. 10. For both the sentiment and the topic, we see that the fixed temperature scheme achieves a higher diversity compared to the decaying temperature scheme.

[1] We refer to `https://github.com/shreyansh26/Extracting-Training-Data-from-Large-Langauge-Models/blob/main/extraction_temperature_decay.py` for an implementation of the temperature decay sampling scheme for HuggingFace models.